

Project 10

Branch-Dependent Planning with the PUMA Positioning Arm

Format: Group of two students; simulation-based project.

Prerequisites: D–H modelling, position inverse kinematics, Jacobians, singularity classification, and path continuation.

Motivation

A wrist-center position may be reached by several shoulder/elbow postures. Those alternatives can have different continuations along the same Cartesian curve. Restricting PUMA to its three-joint positioning task makes the global singularity structure visible while retaining the planning question: which initial solution can follow the complete path?

Task

Overall objectives. Treat PUMA’s first three joints as a noncuspidal 3R wrist-center positioning robot and explain path feasibility through reduced singularity maps and the distribution of its IK solutions.

- **Extract the position task.** Identify the wrist center from the PUMA axis geometry and derive its position as a function of q_1, q_2, q_3 . Validate the D–H and URDF-based descriptions. The wrist joints are irrelevant to this point’s position; do not replace the wrist center with the tool tip.
- **Build the global reference maps.** Derive J_p and plot $\det J_p$ and its zero contours in (q_2, q_3) . Map the singular curves into (ρ, z) and count distinct arm IKs. Identify two-/four-IK regions where present and explain if this specific geometry lacks one of those regions.
- **Select and classify initial solutions.** Choose one regular loop and two translated or scaled variants approaching different arm singularities. Enumerate the initial arm IKs and identify their shoulder/elbow categories. Overlay them on the reduced plot and record mathematical versus joint-limit-admissible configurations.
- **Continue every admissible branch.** Follow each workspace curve from each valid initial arm configuration. Use warm-started continuation and analytical enumeration as a cross-check. Refine the step size near critical events; preserve branch identity instead of replacing a failed solution with an unrelated IK at the next sample.
- **Explain completion and failure.** Compare completion fraction, joint travel, joint-limit margin, and the smallest singular value of J_p . Relate each failed lift to its configuration-space separator and workspace critical curve. A workspace point can have another regular IK even when the selected continuation fails.

Specifics and scope

Use one PUMA positioning model and three position curves, with about 200 samples per curve before local refinement. Keep orientation outside the task and use the same joint limits and tolerances for every comparison. Explore mathematical branch structure first, then report limit filtering. The goal is an explanation of branch-dependent feasibility, not a new general path-search algorithm.

Use the position Jacobian J_p for the 3R task. Confirm that the meridian reduction is valid for the selected model and operational point. Count distinct real IKs modulo periodicity, then filter joint limits separately. Identify two-/four-IK regions where present and explain any absent count rather than forcing both types into every map. Show both determinant values and the zero contour, and distinguish sampled connectivity from a proof. Refine the decisive region instead of claiming completeness from a coarse grid.

Expected outcome

Understand the PUMA arm's D–H model, noncuspidal branch separation, and the difference between workspace reachability and a feasible lift. Deliver reduced determinant maps with overlaid IKs and trajectories, workspace IK-count maps, and a branch-by-branch path comparison.

Submit reproducible code, model parameters and test data, the required plots, a concise 5–6-page report, and a short simulation demonstration. Explain both successful cases and failures; conclusions must follow the numerical and geometric evidence rather than an expected outcome.

Provided materials and implementation

The PUMA 560 robot description (.urdf) and link meshes (.stl) will be provided to the students. Use a robotics library or your own tools for axis/frame inspection, IK visualization, and path animation. Confirm joint offsets and the selected wrist-center frame when importing the model.

Reuse a validated analytical PUMA arm IK implementation, checking every solution against the wrist-center forward model. Implement continuation by seeding each new sample from the preceding configuration and cross-checking with analytical solutions. The wrist joints may be held fixed for visualization because the task constrains only wrist-center position. Use plotting tools to compare branch histories, singularity margins, and completed path fractions.